13. Disorders of Sodium Balance · assets 1-9 / 11

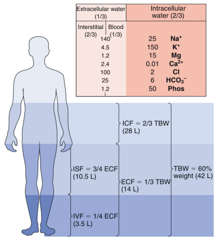

Fig. 13.1 Composition of body fluid compartments. This is a schematic representation of electrolyte composition *(upper panel)* and volumes *(lower panel)* of the body fluid compartments in humans. In the *upper panel*, electrolyte concentrations are in millimoles per liter; intracellular concentrations are typical values obtained from muscle. In the *lower panel, shaded areas* depict the approximate size of each compartment as a function of body weight. In a normally built individual, the total body water content is roughly 60% of body weight. Because adipose tissue has a low concentration of water, the relative water/total body weight ratio is lower in obese individuals. Relative volumes of each compartment are shown as fractions; approximate absolute volumes of the compartments (in liters) in a 70-kg adult are shown in parentheses. *ECF,* Extracellular fluid; *ICF*, intracellular fluid; *ISF*, interstitial fluid; *IVF*, intravascular fluid; *TBW*, total body water. (From Verbalis JG. Body water osmolality. In: Wilkinson B, Jamison R, eds. *Textbook of Nephrology*. London: Chapman & Hall; 1997:89–94. Reproduced with permission of Hodder Arnold.)

Fig_13_1

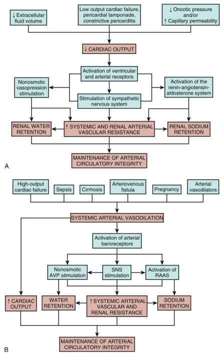

Fig. 13.2 Sensing mechanisms that initiate and maintain renal sodium and water retention in various clinical conditions in which arterial underfilling, with resultant neurohumoral activation and renal sodium and water retention, is caused by a decrease in cardiac output (A) and systemic arterial vasodilation (B). In addition to activating the neurohumoral axis, adrenergic stimulation causes renal vasoconstriction and enhances sodium and fluid transport by the proximal tubule epithelium. (From Schrier RW. Body fluid volume regulation in health and disease: A unifying hypothesis. *Ann Intern Med*. 1990;113:155–159.)

Fig_13_2

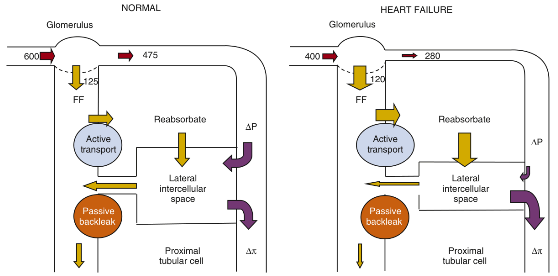

Fig. 13.3 Peritubular control of proximal tubule fluid reabsorption. Fluid reabsorption in the normal state *(left)* and in patients with heart failure *(right)* is shown. Increased postglomerular arteriolar resistance in heart failure is depicted as narrowing. The thickness and font size of the *block arrows* depict relative magnitude of effect. The increase in filtration fraction (FF) in heart failure causes $\Delta\pi$ to rise. The increase in renal vascular resistance in heart failure is believed to reduce ΔP. Both the increase in $\Delta\pi$ and fall in ΔP enhance peritubular capillary uptake of proximal reabsorbate and thus increase absolute Na^+ reabsorption by the proximal tubule. Numbers and *red block arrows* depict renal plasma flow [mL/min] in preglomerular and postglomerular capillaries; ΔP and $\Delta\pi$ are the transcapillary hydraulic and oncotic pressure differences across the peritubular capillary, respectively; *yellow block arrows* indicate transtubular transport; *purple block arrows* represent the effect of peritubular capillary Starling forces on uptake of proximal reabsorbate. (Modified from Humes HD, Gottlieb M, Brenner BM. *The Kidney in Congestive Heart Failure: Contemporary Issues in Nephrology.* Vol 1. New York; Churchill Livingstone; 1978:51–72.)

Fig_13_3

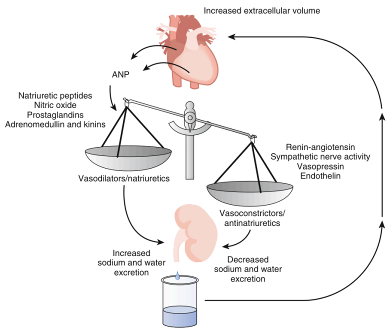

Fig. 13.4 Efferent limb of extracellular fluid volume control in heart failure. Volume homeostasis in heart failure is determined by the balance between natriuretic and antinatriuretic forces. In decompensated heart failure, enhanced activities of the Na^+-retaining systems overwhelm the effects of the vasodilatory/natriuretic systems, which leads to a net reduction in Na^+ excretion and an increase in ECF volume. *ANP,* Atrial natriuretic peptide. (Modified from Winaver J, Hoffman A, Abassi Z, et al. Does the heart's hormone, ANP, help in congestive heart failure? *News Physiol Sci*. 1995;10:247–253.)

Fig_13_4

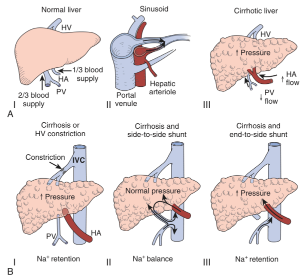

Fig. 13.5 Characteristics of hepatic blood flow. (A) Hepatic circulation. *I,* The normal liver receives two-thirds of its blood flow from the portal vein (PV) and the remaining third from the hepatic artery (HA). *II,* Both the portal venules and hepatic arterioles drain into hepatic sinusoids, but the exact arrangement that allows forward flow of the mixed venous and arterial blood remains unclear. *III,* Cirrhosis increases intrahepatic vascular resistance and sinusoidal pressure. In addition, PV flow is markedly decreased, and HA flow is unchanged or increased. (B) Hepatic vascular hemodynamics and sodium balance. *I,* Cirrhosis or restriction of HV flow increases intrahepatic vascular resistance and sinusoidal pressure, markedly decreasing PV flow and increasing HA flow. Changes in the physical forces or in the composition of the hepatic blood trigger Na^+ retention and edema formation. *II,* Insertion of a side-to-side portocaval shunt decreases sinusoidal pressure and maintains mixing of PV and HA blood, irrigating the liver. Under these conditions and despite cirrhosis, there is no Na^+ retention. *III,* Insertion of an end-to-side portocaval shunt only partially decreases the elevated sinusoidal pressure and prevents mixing of PV and HA blood supplies, inasmuch as the PV blood is diverted to the inferior vena cava (IVC). Under these conditions and, despite normalization of PV pressure, Na^+ retention continues unabated. (Modified from Oliver JA, Verna EC. Afferent mechanisms of sodium retention in cirrhosis and HRS. *Kidney Int*. 2010;77:669–680.)

Fig_13_5

Table 13.1 Mechanisms for Sensing Regional Changes in Effective Arterial Blood Volume

Sensors of Cardiac Filling
Atrial
Neural pathways
Humoral pathways
Ventricular
Pulmonary
Sensors of Cardiac Output
Carotid and aortic baroceptors
Sensors of Organ Perfusion
Renal sensors
CNS sensors
GI tract sensors
Hepatic receptors
Guanylin peptides

CNS, Central nervous system; *GI*, gastrointestinal.

Table_13_1

Table 13.2 Major Renal Effector Mechanisms for Regulating Effective Arterial Blood Volume

Glomerular Filtration Rate and Tubular Reabsorption
Tubuloglomerular feedback
Glomerulotubular balance
Peritubular capillary Starling forces
Luminal composition
Physical factors beyond proximal tubule
Medullary hemodynamics (pressure natriuresis)
Neural Mechanisms
Sympathetic nervous system
Renal nerves
Humoral Mechanisms
Renin-angiotensin-aldosterone system
Vasopressin
Prostaglandins
Natriuretic peptides
Endothelium-derived factors
Endothelins
Nitric oxide
Kinins
Adrenomedullin
Urotensin
Digitalis-like factors
Neuropeptide Y
Apelin
Glucagon-Like Peptide-1
Other novel factors[a]

[a]See Leipziger J, Praetorius H. Renal autocrine and paracrine signaling: a story of self-protection. *Physiol Rev*. 2020;100:1229–1289.

Table_13_2

Table 13.3 Causes of Absolute and Relative Hypovolemia

ABSOLUTE
Extrarenal
Gastrointestinal fluid loss
Bleeding
Skin fluid loss
Respiratory fluid loss
Extracorporeal ultrafiltration
Renal
Diuretics
Obstructive uropathy/postobstructive diuresis
Hormone deficiency
Hypoaldosteronism
Adrenal insufficiency
Na^+ wasting tubulopathies
Genetic
Acquired tubulointerstitial disease
RELATIVE
Extrarenal
Edematous states
Heart failure
Cirrhosis
Generalized vasodilation
Sepsis
Drugs
Pregnancy
Third-space loss
Renal
Severe nephrotic syndrome

Table_13_3

Table 13.4 Causes of Renal Sodium Retention

Primary
Oliguric acute kidney injury
Chronic kidney disease
Glomerular disease
Severe bilateral renal artery stenosis
Na^+-retaining tubulopathies (genetic)
Mineralocorticoid excess
Secondary
Heart failure
Cirrhosis
Idiopathic edema

Table_13_4

Table 13.5 Factors Causing Underfilling of Circulation in Cirrhosis

Peripheral vasodilation and blunted vasoconstrictor response to reflex, chemical, and hormonal influences
Arteriovenous shunts, particularly in portal circulation
Increased vascular capacity of portal and systemic circulation
Hypoalbuminemia
Impaired left ventricular function, so-called cirrhotic cardiomyopathy
Diminished venous return secondary to advanced tense ascites
Occult gastrointestinal bleeding from ulcers, gastritis, or varices
Volume losses caused by vomiting and excessive use of diuretics

Table_13_5

Table 13.6 Renal Effects of Renin-Angiotensin-Aldosterone System Inhibition in Heart Failure

Factors Favoring Improvement in Renal Function
Maintenance of Na^+ balance
Reduction in diuretic dosage
Increase in Na^+ intake
Mean arterial pressure >80 mm Hg
Minimal neurohumoral activation
Intact counterregulatory mechanisms
Factors Favoring Deterioration in Renal Function
Evidence of Na^+ depletion or poor renal perfusion
Large doses of diuretics
Increased urea/creatinine ratio
Mean arterial pressure <80 mm Hg
Evidence of maximal neurohumoral activation
AVP-induced hyponatremia
Interruption of counterregulatory mechanisms
Coadministration of prostaglandin inhibitors
Adrenergic dysfunction (e.g., diabetes mellitus)

AVP, Arginine vasopressin.

Table_13_6